Math **163** Name (PLEASE PRINT) Captain Yellowbeard

Exam **2A** **2024** Fall/Spring/**Summer I** /Summer II Score _____

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	3	3								6	
1b	3	3								6	
1c											
2a	6									6	
2b	6									6	
2c											
3a	2	4								6	
3b	6									6	
3c											
								Total		36	

Instructions:

Your J-number = 00407205

$a := mnm_{k \in 1..8}\{J_k | (k \in 1..8) \wedge (J_k \neq 0)\} =$ 2

$b := mxm(J_k | k \in 1..8) =$ 7

$c := \left\lfloor \left(\frac{b+a}{2}\right) \right. = \left\lfloor \left(\frac{7+2}{2}\right) \right. = \left\lfloor \left(\frac{9}{2}\right) \right. = \left\lfloor \left((4)+\left(\frac{1}{2}\right)\right) \right. =$ 4 =

$d :=$ the smallest **odd** number **strictly greater** than c = 5

6 (1a) Expand using the binomial theorem **exactly as in the Notes.**

3 (i) $(b - ax)^c$

My problem: $\left((7) - (2)x\right)^{(4)}$

$= \left((7) + (-2)x\right)^{(4)}$

$= 2401$

$+ (-1)2744x$

$+ 1176x^2$

$+ (-1)224x^3$

$+ 16x^4$

1a.i.0

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

$$(7+(-2)x)^4 = \sum_{k=0}^{4} \binom{4}{k} 7^{4-k} ((-2)x)^k$$

$\langle n \longleftarrow 4,\ a \longleftarrow 7,\ b \longleftarrow ((-2)x) \rangle$

1a.i.1

$$\sum_{k=0}^{4} \binom{4}{k} 7^{4-k} ((-2)x)^k$$

$$= \binom{4}{0} 7^{4-0} ((-2)x)^0 + \binom{4}{1} 7^{4-1} ((-2)x)^1 + \binom{4}{2} 7^{4-2} ((-2)x)^2 + \binom{4}{3} 7^{4-3} ((-2)x)^3 + \binom{4}{4} 7^{4-4} ((-2)x)^4$$

$$= 2401 + (-1)2744x + 1176xx + (-1)224xxx + 16xxxx$$

$$= 2401 + (-1)2744x + 1176x^2 + (-1)224x^3 + 16x^4$$

1a.i.2

$$\binom{4}{4} = \binom{4}{4-4} = \binom{4}{0} = 1$$

1a.i.3

$$\binom{4}{3} = \binom{4}{4-3} = \binom{4}{1} = 4$$

1a.i.4

$$\binom{4}{2} = \binom{4}{4-2} = \binom{4}{2} = \frac{4!}{(2!)((4-2)!)} = \frac{4!}{(2!)(2!)} = \frac{(4)(3)\cancel{(2)(1)}}{\cancel{(2)(1)}(2)(1)} = \frac{(4)(3)}{(2)(1)} = \frac{12}{2} = 6$$

1a.i.5

$$\binom{4}{0} 7^{4-0} ((-2)x)^0 = (1)(2401)(1) = 2401$$

1a.i.6

$$\binom{4}{1} 7^{4-1} ((-2)x)^1 = (4)7^3((-2)x) = (-1)2744x$$

1a.i.7

$$\binom{4}{2} 7^{4-2} ((-2)x)^2 = (6)7^2(4)xx = 1176xx$$

1a.i.8

$$\binom{4}{3} 7^{4-3} ((-2)x)^3 = (4)7^1(-8)xxx = (-1)224xxx$$

1a.i.9

$$\binom{4}{4} 7^{4-4} ((-2)x)^4 = (1)7^0(16)xxxx = 16xxxx$$

(1a) Expand using the multinomial theorem **exactly as in the Notes.**

3 (ii) $(c - x + y)^{1+a}$

My problem: $\left((4) - (1)x + y\right)^{1+(2)}$

$$= \left((4) + (-1)x + y\right)^{(3)}$$

$$= 64 + (-1)48x + 12x^2 + (-1)x^3 + (-1)24xy + 3x^2y + (-1)3xy^2 + 48y + 12y^2 + y^3$$

1a.ii.0

$$(x_1 + x_2 + \ldots + x_p)^n = \sum_{n_1+n_2+\ldots+n_p=n} \binom{n}{n_1\ n_2 \ldots n_p} (x_1^{n_1})(x_2^{n_2}) \ldots (x_p^{n_p})$$

$$(4 + (-1)x + y)^3 = \sum_{n_1+n_2+n_3=3} \binom{3}{n_1\ n_2\ n_3} (4^{n_1})(((-1)x)^{n_2})(y^{n_3})$$

$\langle p \leftarrow 3,\ n \leftarrow 3,\ x_1 \leftarrow 4,\ x_2 \leftarrow (-1)x,\ x_3 \leftarrow y \rangle$

1a.ii.1

$$\begin{aligned}
3 &= 0+0+3 \\
&= 0+1+2 \\
&= 0+2+1 \\
&= 0+3+0 \\
&= 1+0+2 \\
&= 1+1+1 \\
&= 1+2+0 \\
&= 2+0+1 \\
&= 2+1+0 \\
&= 3+0+0
\end{aligned}$$

$$\sum_{n_1+n_2+n_3=3} \binom{3}{n_1\ n_2\ n_3} (4^{n_1})(((-1)x)^{n_2})(y^{n_3})$$

$= \binom{3}{0\,0\,3} 4^0((-1)x)^0 y^3$	$= yyy$	$= 64$	$= 64$
$+ \binom{3}{0\,1\,2} 4^0((-1)x)^1 y^2$	$+ (-1)3xyy$	$+ (-1)48x$	$+ (-1)48x$
$+ \binom{3}{0\,2\,1} 4^0((-1)x)^2 y^1$	$+ 3xxy$	$+ 12xx$	$+ 12x^2$
$+ \binom{3}{0\,3\,0} 4^0((-1)x)^3 y^0$	$+ (-1)xxx$	$+ (-1)xxx$	$+ (-1)x^3$
$+ \binom{3}{1\,0\,2} 4^1((-1)x)^0 y^2$	$+ 12yy$	$+ (-1)24xy$	$+ (-1)24xy$
$+ \binom{3}{1\,1\,1} 4^1((-1)x)^1 y^1$	$+ (-1)24xy$	$+ 3xxy$	$+ 3x^2y$
$+ \binom{3}{1\,2\,0} 4^1((-1)x)^2 y^0$	$+ 12xx$	$+ (-1)3xyy$	$+ (-1)3xy^2$
$+ \binom{3}{2\,0\,1} 4^2((-1)x)^0 y^1$	$+ 48y$	$+ 48y$	$+ 48y$
$+ \binom{3}{2\,1\,0} 4^2((-1)x)^1 y^0$	$+ (-1)48x$	$+ 12yy$	$+ 12y^2$
$+ \binom{3}{3\,0\,0} 4^3((-1)x)^0 y^0$	$+ 64$	$+ yyy$	$+ y^3$

1a.ii.3

$$\begin{aligned}
&\binom{3}{0\,0\,3} \\
&= \binom{3}{0\,3\,0} \\
&= \binom{3}{3\,0\,0} \\
&= \frac{(3!)}{(3!)(0!)(0!)} \\
&= \frac{\cancel{(3)}\cancel{(2)}(1)}{\cancel{(3)}\cancel{(2)}(1)(1)(1)} \\
&= 1
\end{aligned}$$

1a.ii.4

$$\begin{aligned}
&\binom{3}{0\,1\,2} \\
&= \binom{3}{0\,2\,1} \\
&= \binom{3}{1\,0\,2} \\
&= \binom{3}{1\,2\,0} \\
&= \binom{3}{2\,0\,1} \\
&= \binom{3}{2\,1\,0} \\
&= \frac{(3!)}{(2!)(1!)(0!)} \\
&= \frac{(3)\cancel{(2)}(1)}{\cancel{(2)}(1)(1)(1)} \\
&= 3
\end{aligned}$$

1a.ii.5

$$\begin{aligned}
&\binom{3}{1\,1\,1} \\
&= \frac{(3!)}{(1!)(1!)(1!)} \\
&= \frac{(3)(2)(1)}{(1)(1)(1)} \\
&= 6
\end{aligned}$$

1a.ii.6

$$\binom{3}{0\,0\,3} 4^0((-1)x)^0 y^3$$
$$= (1)(1)(1)yyy$$
$$= yyy$$

1a.ii.7

$$\binom{3}{0\,1\,2} 4^0((-1)x)^1 y^2$$
$$= (3)(1)(-1)xyy$$
$$= (-1)3xyy$$

1a.ii.8

$$\binom{3}{0\,2\,1} 4^0((-1)x)^2 y^1$$
$$= (3)(1)(1)xxy$$
$$= 3xxy$$

1a.ii.9

$$\binom{3}{0\,3\,0} 4^0((-1)x)^3 y^0$$
$$= (1)(1)(-1)xxx(1)$$
$$= (-1)xxx$$

1a.ii.10

$$\binom{3}{1\,0\,2} 4^1((-1)x)^0 y^2$$
$$= (3)(4)(1)yy$$
$$= 12yy$$

1a.ii.11

$$\binom{3}{1\,1\,1} 4^1((-1)x)^1 y^1$$
$$= (6)(4)(-1)xy$$
$$= (-1)24xy$$

1a.ii.12

$$\binom{3}{1\,2\,0} 4^1((-1)x)^2 y^0$$
$$= (3)(4)(1)xx(1)$$
$$= 12xx$$

1a.ii.13

$$\binom{3}{2\,0\,1} 4^2((-1)x)^0 y^1$$
$$= (3)(16)(1)y$$
$$= 48y$$

1a.ii.14

$$\binom{3}{2\,1\,0} 4^2((-1)x)^1 y^0$$
$$= (3)(16)(-1)x(1)$$
$$= (-1)48x$$

1a.ii.15

$$\binom{3}{3\,0\,0} 4^3((-1)x)^0 y^0$$
$$= (1)(64)(1)(1)$$
$$= 64$$

6 (1b) Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences in the Notes.

3 (i)
$$\begin{cases} \rho(0) \coloneqq 1 \\ \rho(1) \coloneqq 2 \\ \forall n \in \mathbb{N} \quad \Big(\ \rho(n+2)\Big) - (b+c)\Big(\rho(n+1)\Big) + (bc)\Big(\rho(n)\Big) = 0 \end{cases}$$

First rewrite the recurrence with your values of ***b*** **and** ***c***.

My problem:

$$\Big(\ \rho(n+2)\Big) - \Big((7)+(4)\Big)\Big(\rho(n+1)\Big) + \Big((7)(4)\Big)\Big(\rho(n)\Big) = 0$$

My problem:

$$\Big(\ |\ \Big)\Big(\ \rho(n+2)\Big) + \Big(-11\Big)\Big(\rho(n+1)\Big) + (28)\Big(\rho(n)\Big) = 0$$

My solution: $\forall n \in \mathbb{N}$

$$\rho(n)$$

$$= \quad \tfrac{5}{3}(4)^n - \tfrac{2}{3}(7)^n$$

16.i.0

$$\rho(n+2) - 11\rho(n+1) + 28\rho(n) = 0$$
$$- \left(-11\rho(n+1) + 28\rho(n)\right) \quad - \left(-11\rho(n+1) + 28\rho(n)\right)$$
$$(1)(\rho(n+2)) = -(-11)(\rho(n+1)) - (28)(\rho(n))$$

16.i.1

$$(1)x^2 + (-11)x + 28 = 0$$
$$(x-7)(x-4) = 0$$
$$(x-7=0) \vee (x-4=0)$$
$$(x=7) \vee (x=4)$$
$$(r_1=4) \wedge (r_2=7)$$
$$r_1 \neq r_2$$
$$\forall n \in \mathbb{N} \quad \rho(n) = k_1(4)^n + k_2(7)^n$$

$$\forall n \in \mathbb{N} \quad \rho(n) = \left(\tfrac{5}{3}\right)(4)^n + (-1)\left(\tfrac{2}{3}\right)(7)^n = \tfrac{5}{3}(4)^n - \tfrac{2}{3}(7)^n$$

16.i.2

$$\frac{\rho(n) = k_1(4)^n + k_2(7)^n}{\rho(0) = k_1(4)^0 + k_2(7)^0} \langle n \leftarrow 0 \rangle \qquad \rho(0) := 1$$
$$1 = k_1(1) + k_2(1)$$

$$\frac{\rho(n) = k_1(4)^n + k_2(7)^n}{\rho(1) = k_1(4)^1 + k_2(7)^1} \langle n \leftarrow 1 \rangle \qquad \rho(1) := 2$$
$$2 = k_1(4) + k_2(7)$$

16.i.3

$$\frac{1 = k_1(1) + k_2(1)}{k_2 = 1 - k_1} \qquad \frac{2 = k_1(4) + k_2(7)}{k_1(4) = 2 - k_2(7)}$$
$$k_1(4) = 2 - (1 - k_1)(7)$$
$$k_1(4) = 2 - 7 + k_1(7)$$
$$k_1(4) = -5 + k_1(7)$$
$$-k_1(4) \qquad -k_1(4)$$
$$0 = -5 + k_1(3)$$
$$k_1(3) = 5$$
$$\div 3 \quad \div 3$$
$$k_1 = \tfrac{5}{3}$$
$$k_2 = 1 - k_1 = 1 - \tfrac{5}{3} = -\tfrac{2}{3}$$

(1b) Continued. Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences given in the Notes, using substitution. Prove by substitution that your answer is correct.

3 (ii)
$$\begin{cases} \rho(0) \coloneqq 1 \\ \rho(1) \coloneqq 2 \\ \forall n \in \mathbb{N} \quad (a^2)\Big(\ \rho(n+2)\Big) - (2ac)\Big(\rho(n+1)\Big) + (c^2)\Big(\rho(n)\Big) = 0 \end{cases}$$

First rewrite with your values of ***a*** **and** ***c***.

My problem:

$$\Big((2)^2\Big)\Big(\ \rho(n+2)\Big) - \Big(2(2)(4)\Big)\Big(\rho(n+1)\Big) + \Big((4)^2\Big)\Big(\rho(n)\Big) = 0$$

My problem:

$$\Big(4\Big)\Big(\ \rho(n+2)\Big) + \Big(-16\quad\Big)\Big(\rho(n+1)\Big) + (16)\Big(\rho(n)\Big) = 0$$

My solution: $\forall n \in \mathbb{N}$

$$\rho(n)$$

$$= \quad 2^n$$

1b.ii.0

$$(4)(\rho(n+2)) + (-16)(\rho(n+1)) + (16)(\rho(n)) = 0$$

$$-\big((-16)(\rho(n+1)) + (16)(\rho(n))\big) \qquad -\big((-16)(\rho(n+1)) + (16)(\rho(n))\big)$$

$$(4)(\rho(n+2)) = (16)(\rho(n+1)) - (16)(\rho(n))$$

$$\div 4 \qquad\qquad \div 4$$

$$(1)(\rho(n+2)) = (4)(\rho(n+1)) - (4)(\rho(n))$$

1b.ii.1

$$(4)x^2 + (-16)x + 16 = 0$$

$$4(x-2)^2 = 0$$

$$x = 2$$

$$r_1 = x = r_2$$

$$\forall n \in \mathbb{N} \quad \rho(n) = k_1((2)^n) + k_2(n(2)^n)$$

$$\forall n \in \mathbb{N} \quad \rho(n) = (1)((2)^n) + (0)(n(2)^n) = 2^n$$

1b.ii.2

$$\rho(n) = k_1((2)^n) + k_2(n(2)^n) \quad \langle n \leftarrow 0 \rangle$$

$$\rho(0) = k_1((2)^0) + k_2((0)(2)^0) = k_1(1) + (0) = k_1 \qquad \rho(0) := 1$$

$$k_1 = 1$$

1b.ii.3

$$\rho(n) = k_1((2)^n) + k_2(n(2)^n) \quad \langle n \leftarrow 1 \rangle$$

$$\rho(1) = k_1((2)^1) + k_2((1)(2)^1) = k_1(2) + k_2(2) \qquad \rho(1) := 2$$

$$2 = k_1(2) + k_2(2)$$

$$\div 2 \qquad\qquad \div 2$$

$$1 = k_1 + k_2$$

1b.ii.4

$$k_1 = 1$$

$$k_1 + k_2 = 1$$

$$1 + k_2 = 1$$

$$-1 \qquad -1$$

$$k_2 = 0$$

1b.ii.5

$$\forall n \in \mathbb{N} \left(\begin{array}{l} \rho(n) \\ := \left(\rho(n) = 2^{(n)}\right) \end{array} \right)$$

1b.ii.6

$$\frac{\rho(n) := (\rho(n) = 2^{(n)})}{\rho(0) := (\rho(0) = 2^{(0)})} \langle n \leftarrow 0 \rangle$$

1b.ii.7

$$\frac{\rho(n) := (\rho(n) = 2^{(n)})}{\rho(1) := (\rho(1) = 2^{(1)})} \langle n \leftarrow 1 \rangle$$

1b.ii.8

BC ⓪

$$\rho(0) := 1 \qquad 2^0 = 1$$

$$\rho(0) = 1 = 2^0$$

$$\rho(0) = 2^0$$

$$\rho(0)$$

1b.ii.9

BC ①

$$\rho(1) := 2 \qquad 2^1 = 2$$

$$\rho(1) = 2 = 2^1$$

$$\rho(1) = 2^1$$

$$\rho(1)$$

1b.ii.10

$$\frac{\rho(n) := (\rho(n) = 2^{(n)})}{\rho((n+2)) := (\rho((n+2)) = 2^{((n+2))})} \langle n \leftarrow (n+2) \rangle$$

1b.ii.11

RcS

$$\rho(0) \quad \rho(1) \quad \dots \quad \rho(n) \quad \rho(n+1)$$

$$\rho(n+2)$$

$$= (4)(\rho(n+1)) - (4)(\rho(n))$$

$$= (4)(\rho(n+1) - \rho(n))$$

$$= (4)(2^{(n+1)} - 2^n)$$

$$= (4)(2^n)((2)-(1))$$

$$= (4)(2^n)(1)$$

$$= (2^2)(2^n)(1)$$

$$= 2^{(n+2)}$$

$$\rho(n+2) = 2^{(n+2)}$$

$$\rho(n+2)$$

1b.ii.12

$$\frac{\rho(0) \quad \left(\frac{\rho(0) \ \rho(1) \dots \rho(n)}{\rho(n+1)}\right)}{\forall n \in \mathbb{N} \ (\rho(n))}$$

$$\forall n \in \mathbb{N} \ (\rho(n) = 2^n)$$

6 (2a) Consider the following recurrence relation for $\varphi(n)$ and $\alpha(n)$ and **answer the questions**:

$$\begin{cases} & \alpha(0) \coloneqq a \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)b \end{cases}$$

$$\begin{cases} & \varphi(0) \coloneqq \alpha(0) + c \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)d + a \end{cases}$$

Show calculations on pages that follow, numbered according to instructions.

Rewrite the recursion using your values of a, b, c, and d.

My problem:

$$\begin{cases} & \alpha(0) \coloneqq (2) \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)(7) \end{cases} \qquad \begin{cases} & \varphi(0) \coloneqq \alpha(0) + c = (2) + (4) \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)(5) + (2) \end{cases}$$

Show full computations on pages that follow, numbered as per instructions, for all values and your conjecture:

$\alpha(0) = 2$ $\varphi(0) = 6$

$\alpha(1) = 42$ $\varphi(1) = 12$

$\alpha(2) = 84$ $\varphi(2) = 212$

$\alpha(3) = 1484$ $\varphi(3) = 422$

Conjecture : $\forall n \in \mathbb{N}$

$\alpha(n) =$ see following page $\varphi(n) =$ see following page

2a.0

$$\begin{array}{rcl} \alpha(n+1) & = & (\varphi(n))(7) \\ \hline \alpha(0+1) & = & (\varphi(0))(7) \end{array} \langle n \leftarrow 0 \rangle$$

$$\begin{aligned} \alpha(1) &= (6)(7) \\ &= 42 \end{aligned}$$

2a.1

$$\begin{array}{rcl} \varphi(n+1) & = & (\alpha(n))(5)+2 \\ \hline \varphi(0+1) & = & (\alpha(0))(5)+2 \end{array} \langle n \leftarrow 0 \rangle$$

$$\begin{aligned} \varphi(1) &= (2)(5)+2 \\ &= 10+2 \\ &= 12 \end{aligned}$$

2a.2

$$\begin{array}{rcl} \alpha(n+1) & = & (\varphi(n))(7) \\ \hline \alpha(1+1) & = & (\varphi(1))(7) \end{array} \langle n \leftarrow 1 \rangle$$

$$\begin{aligned} \alpha(2) &= (12)(7) \\ &= 84 \end{aligned}$$

2a.3

$$\begin{array}{rcl} \varphi(n+1) & = & (\alpha(n))(5)+2 \\ \hline \varphi(1+1) & = & (\alpha(1))(5)+2 \end{array} \langle n \leftarrow 1 \rangle$$

$$\begin{aligned} \varphi(2) &= (42)(5)+2 \\ &= 210+2 \\ &= 212 \end{aligned}$$

2a.4

$$\begin{array}{rcl} \alpha(n+1) & = & (\varphi(n))(7) \\ \hline \alpha(2+1) & = & (\varphi(2))(7) \end{array} \langle n \leftarrow 2 \rangle$$

$$\begin{aligned} \alpha(3) &= (212)(7) \\ &= 1484 \end{aligned}$$

2a.5

$$\begin{array}{rcl} \varphi(n+1) & = & (\alpha(n))(5)+2 \\ \hline \varphi(2+1) & = & (\alpha(2))(5)+2 \end{array} \langle n \leftarrow 2 \rangle$$

$$\begin{aligned} \varphi(3) &= (84)(5)+2 \\ &= 420+2 \\ &= 422 \end{aligned}$$

2a.6

$$\begin{aligned} \alpha(n+1) &:= (\varphi(n))(7) \\ &= ((\alpha(n-1))(5)+2))(7) \\ &= (35)(\alpha(n-1))+14 \end{aligned}$$

2a.7

$$\begin{aligned} \varphi(n+1) &:= (\alpha(n))(5)+2 \\ &= ((\varphi(n-1))(7)(5)+2 \\ &= (35)((\varphi(n-1))+2 \end{aligned}$$

2a.8

$$\begin{aligned} & \sum_{k=0}^{\left(\left(\left|\frac{n}{2}\right|\right)-1\right)} \left((35)^k\right) \\ &= Sm\left((35)^n, \left(\left(\left|\frac{n}{2}\right|\right)-1\right)\right) \\ &= \frac{1-(35)^{\left(\left(\left(\left|\frac{n}{2}\right|\right)-1\right)+1\right)}}{1-(35)} \\ &= \frac{(35)^{\left(\left|\frac{n}{2}\right|\right)}-1}{34} \end{aligned}$$

2a.9

$$\begin{aligned} \alpha(0) &= 2 \\ \alpha(1) &= 42 \\ \alpha(2) &= (35)(2)+14 \\ \alpha(3) &= (35)(42)+14 \\ \alpha(4) &= (35)((35)(2)+14)+14 = (35)^2(2)+(35)(14)+14 \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

$$\alpha(n)$$

$$= (35)^{\left(\left|\frac{n}{2}\right|\right)}\left((2)\left(\frac{1+(-1)^n}{2}\right) + (42)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (14)\left(\sum_{k=0}^{\left(\left(\left|\frac{n}{2}\right|\right)-1\right)} (35)^k\right)$$

$$= (35)^{\left(\left|\frac{n}{2}\right|\right)}\left((2)\left(\frac{1+(-1)^n}{2}\right) + (42)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (14)\left(\frac{(35)^{\left(\left|\frac{n}{2}\right|\right)}-1}{34}\right)$$

2a.10

$$\begin{aligned} \varphi(0) &= 6 \\ \varphi(1) &= 12 \\ \varphi(2) &= (35)(6)+2 \\ \varphi(3) &= (35)(12)+2 \\ \varphi(4) &= (35)((35)(6)+2)+2 = (35)^2(6)+(35)(2)+2 \\ &\vdots \qquad\qquad \vdots \end{aligned}$$

$$\varphi(n)$$

$$= (35)^{\left(\left|\frac{n}{2}\right|\right)}\left((6)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (2)\left(\sum_{k=0}^{\left(\left(\left|\frac{n}{2}\right|\right)-1\right)} (35)^k\right)$$

$$= (35)^{\left(\left|\frac{n}{2}\right|\right)}\left((6)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (2)\left(\frac{(35)^{\left(\left|\frac{n}{2}\right|\right)}-1}{34}\right)$$

6 (2b) Find conjectures for the sums:

$$Sm(\alpha, n) \coloneqq S(\alpha, n) \coloneqq \sum_{k=0}^{n} \alpha(k) \quad \text{and}$$

$$Sm(\varphi, n) \coloneqq S(\varphi, n) \coloneqq \sum_{k=0}^{n} \varphi(k)$$

where $\alpha(n)$ and $\varphi(n)$ are defined, as in (2a) by the recursion appearing below,

$$\begin{cases} \boldsymbol{\alpha(0) \coloneqq a} \\ \forall \boldsymbol{n} \in \mathbb{N} \quad \boldsymbol{\alpha(n+1) \coloneqq (\varphi(n))b} \end{cases} \qquad \begin{cases} \boldsymbol{\varphi(0) \coloneqq \alpha(0) + c} \\ \forall \boldsymbol{n} \in \mathbb{N} \quad \boldsymbol{\varphi(n+1) \coloneqq (\alpha(n))d + a} \end{cases}$$

Show computations on pages that follow, numbered according to instructions, to indicate how you arrive at your conjecture.

Rewrite the recursion using your values of a, b, and c.

My problem:

$$\begin{cases} \alpha(0) \coloneqq (2) \\ \forall n \in \mathbb{N} \quad \alpha(n+1) \coloneqq (\varphi(n))(7) \end{cases} \qquad \begin{cases} \varphi(0) \coloneqq \alpha(0) + (4) = (2) + (4) \\ \forall n \in \mathbb{N} \quad \varphi(n+1) \coloneqq (\alpha(n))(5) + (2) \end{cases}$$

My Conjectures :

(1) $\forall n \in \mathbb{N}$ $Sm(\alpha, n)$

$$= \left(\frac{41}{17}\right)\left(\frac{(35)^{\left(\left(\left\lfloor\left(\frac{n}{2}\right)\right\rfloor\right)+1\right)} - 1}{34}\right) + \left(\frac{721}{17}\right)\left(\frac{(35)^{\left(\left(\left\lfloor\left(\frac{n-1}{2}\right)\right\rfloor\right)+1\right)} - 1}{34}\right) - \left(\frac{7}{17}\right)(n+1)$$

(2) $\forall n \in \mathbb{N}$ $Sm(\varphi, n)$

$$= \left(\frac{103}{17}\right)\left(\frac{(35)^{\left(\left(\left\lfloor\left(\frac{n}{2}\right)\right\rfloor\right)+1\right)} - 1}{34}\right) + \left(\frac{205}{17}\right)\left(\frac{(35)^{\left(\left(\left\lfloor\left(\frac{n-1}{2}\right)\right\rfloor\right)+1\right)} - 1}{34}\right) - \left(\frac{1}{34}\right)(n+1)$$

$$\begin{aligned}
\alpha(0) &= 2 \\
\alpha(1) &= 42 \\
\alpha(2) &= (35)(2)+14 \\
\alpha(3) &= (35)(42)+14 \\
\alpha(4) &= (35)((35)(2)+14)+14 = (35)^2(2)+(35)(14)+14
\end{aligned}$$

$$\begin{aligned}
Sm(\alpha,4) &= (2)(1+35+(35)^2) + (14)((0)+(1)+(1+35)) \\
&\quad + (42)(1+35) + (14)((0)+(1)) \\
&= (2)\textstyle\sum_{k=0}^{2}((35)^k) + (14)\left(\sum_{k=0}^{-1}((35)^k) + \sum_{k=0}^{0}((35)^k) + \sum_{k=0}^{1}((35)^k)\right) \\
&\quad + (42)\textstyle\sum_{k=0}^{2-1}((35)^k) + (14)\left(\sum_{k=0}^{-1}((35)^k) + \sum_{k=0}^{0}((35)^k)\right)
\end{aligned}$$

$$\begin{aligned}
&Sm(\alpha,4) \\
&= \left((2)(Sm(35,2)) + (14)\left(Sm(35,-1)+Sm(35,0)+Sm(35,1)\right)\right) + \left((42)(Sm(35,2-1)) + (14)\left(Sm(35,-1)+Sm(35,0)\right)\right) \\
&= \left((2)(Sm(35,2)) + (14)\left(\tfrac{(35)^0-1}{35-1} + \tfrac{(35)^1-1}{35-1} + \tfrac{(35)^2-1}{35-1}\right)\right) + \left((42)(Sm(35,2-1)) + (14)\left(\tfrac{(35)^0-1}{35-1} + \tfrac{(35)^1-1}{35-1}\right)\right) \\
&= \left((2)(Sm(35,2)) + \left(\tfrac{14}{35-1}\right)\left((35)^0-1)+((35)^1-1)+((35)^2-1)\right)\right) + \left((42)(Sm(35,2-1)) + \left(\tfrac{14}{34}\right)\left((35)^0-1)+((35)^1-1)\right)\right) \\
&= \left((2)(Sm(35,2)) + \left(\tfrac{14}{34}\right)\left((35)^0+(35)^1+(35)^2)-(1+1+1)\right)\right) + \left((42)(Sm(35,2-1)) + \left(\tfrac{14}{34}\right)\left((35)^0+(35)^1)-(1+1)\right)\right) \\
&= \left((2)(Sm(35,2)) + \left(\tfrac{14}{34}\right)\left(\left(\textstyle\sum_{k=0}^{2}((35)^k)\right)-(1)(2+1)\right)\right) + \left((42)(Sm(35,2-1)) + \left(\tfrac{14}{34}\right)\left(\left(\textstyle\sum_{k=0}^{2-1}((35)^k)\right)-(1)(1+1)\right)\right)
\end{aligned}$$

$$\begin{aligned}
&Sm(\alpha,n) \\
&= \left((2)(Sm(35,\lfloor(\tfrac{n}{2})\rfloor)) + \left(\tfrac{14}{34}\right)\left((Sm(35,\lfloor(\tfrac{n}{2})\rfloor)-(\lfloor(\tfrac{n}{2})\rfloor-1)\right) + \left((42)(Sm(35,\lfloor\tfrac{(n-1)}{2}\rfloor)) + \left(\tfrac{14}{34}\right)\left((Sm(35,\lfloor\tfrac{(n-1)}{2}\rfloor)-(\lfloor\tfrac{(n-1)}{2}\rfloor-1)\right)\right) \\
&= \left((2)+\left(\tfrac{14}{34}\right)\right)(Sm(35,\lfloor(\tfrac{n}{2})\rfloor)) + \left((42)+\left(\tfrac{14}{34}\right)\right)(Sm(35,\lfloor\tfrac{(n-1)}{2}\rfloor)) - \left(\tfrac{14}{34}\right)\left(\left(\lfloor\tfrac{(n-1)}{2}\rfloor\right)+1+\left(\lfloor(\tfrac{n}{2})\rfloor+1\right)\right) \\
&= \left(\tfrac{82}{34}\right)(Sm(35,\lfloor(\tfrac{n}{2})\rfloor)) + \left(\tfrac{1442}{34}\right)(Sm(35,\lfloor\tfrac{(n-1)}{2}\rfloor)) - \left(\tfrac{14}{34}\right)((n-1)+2) \\
&= \left(\tfrac{41}{17}\right)\left(\frac{(35)^{\left(\left(\lfloor(\frac{n}{2})\rfloor\right)+1\right)}-1}{34}\right) + \left(\tfrac{721}{17}\right)\left(\frac{(35)^{\left(\left(\lfloor\frac{(n-1)}{2}\rfloor\right)+1\right)}-1}{34}\right) - \left(\tfrac{7}{17}\right)(n+1)
\end{aligned}$$

1

$$\begin{aligned}
&\textstyle\sum_{k=0}^{-1}((35)^k) \\
&= Sm((35)^k,-1) \\
&= \frac{1-(35)^{((-1)+1)}}{1-(35)} \\
&= \frac{(35)^0-1}{34} \\
&= \frac{1-1}{34} \\
&= \frac{0}{34} \\
&= 0
\end{aligned}$$

2b.ii.0

$$\forall p \in \mathbb{N}(\mathrm{Evn}(p))$$

$$\varphi(n) = (35)^{\left(\left(\frac{n}{2}\right)\right)}\left((6)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (2)\left(\frac{(35)^{\left(\left(\frac{n}{2}\right)\right)} - 1}{34}\right) \quad \langle n \leftarrow p \rangle$$

$$\varphi(p) = (35)^{\left(\left(\frac{p}{2}\right)\right)}\left((6)\left(\frac{1+(-1)^p}{2}\right) + (12)\left(\frac{1+(-1)^{(p+1)}}{2}\right)\right) + (2)\left(\frac{(35)^{\left(\left(\frac{p}{2}\right)\right)} - 1}{34}\right)$$

$$= (35)^{\frac{p}{2}}\left((6)(1) + (12)(0)\right) + (2)\left(\frac{(35)^{\frac{p}{2}} - 1}{34}\right)$$

$$= (35)^{\frac{p}{2}}(6) + \left(\frac{1}{17}\right)(35)^{\frac{p}{2}} - \left(\frac{1}{34}\right)$$

1

$$\forall q \in \mathbb{N}(\mathrm{Odd}(q))$$

$$\varphi(n) = (35)^{\left(\left(\frac{n}{2}\right)\right)}\left((6)\left(\frac{1+(-1)^n}{2}\right) + (12)\left(\frac{1+(-1)^{(n+1)}}{2}\right)\right) + (2)\left(\frac{(35)^{\left(\left(\frac{n}{2}\right)\right)} - 1}{34}\right) \quad \langle n \leftarrow q \rangle$$

$$\varphi(q) = (35)^{\left(\left(\frac{q}{2}\right)\right)}\left((6)\left(\frac{1+(-1)^q}{2}\right) + (12)\left(\frac{1+(-1)^{(q+1)}}{2}\right)\right) + (2)\left(\frac{(35)^{\left(\left(\frac{q}{2}\right)\right)} - 1}{34}\right)$$

$$= (35)^{\frac{(q-1)}{2}}\left((6)(0) + (12)(1)\right) + (2)\left(\frac{(35)^{\frac{(q-1)}{2}} - 1}{34}\right)$$

$$= (35)^{\frac{(q-1)}{2}}(12) + \left(\frac{1}{17}\right)(35)^{\frac{(q-1)}{2}} - \left(\frac{1}{34}\right)$$

2

$$\frac{\sum_{k=0}^{n}\left((35)^k\right) = \frac{(35)^{n+1}-1}{34}}{\sum_{k=0}^{\left(\left(\frac{n}{2}\right)\right)}\left((35)^k\right) = \left(\frac{(35)^{\left(\left(\left(\frac{n}{2}\right)\right)+1\right)}-1}{34}\right)} \quad \left\langle n \leftarrow \left(\left(\frac{n}{2}\right)\right)\right\rangle$$

3

$$\frac{\sum_{k=0}^{n}\left((35)^k\right) = \frac{(35)^{n+1}-1}{34}}{\sum_{k=0}^{\left(\left(\frac{(n-1)}{2}\right)\right)}\left((35)^k\right) = \left(\frac{(35)^{\left(\left(\left(\frac{(n-1)}{2}\right)\right)+1\right)}-1}{34}\right)} \quad \left\langle n \leftarrow \left(\left(\frac{(n-1)}{2}\right)\right)\right\rangle$$

4

$$\sum_{k=0}^{n}\left((35)^k\right)$$

$$= Sm((35)^k, n)$$

$$= \frac{1-(35)^{(n+1)}}{1-(35)}$$

$$= \frac{(35)^{(n+1)} - 1}{34}$$

5

$$\varphi(0) = (35)^{\frac{0}{2}}(6) + \left(\frac{1}{17}\right)(35)^{\frac{0}{2}} - \left(\frac{1}{34}\right)$$

$$\varphi(1) = (35)^{\frac{(1-1)}{2}}(12) + \left(\frac{1}{17}\right)(35)^{\frac{(1-1)}{2}} - \left(\frac{1}{34}\right)$$

$$\varphi(2) = (35)^{\frac{2}{2}}(6) + \left(\frac{1}{17}\right)(35)^{\frac{2}{2}} - \left(\frac{1}{34}\right)$$

$$\varphi(3) = (35)^{\frac{(3-1)}{2}}(12) + \left(\frac{1}{17}\right)(35)^{\frac{(3-1)}{2}} - \left(\frac{1}{34}\right)$$

$$\varphi(4) = (35)^{\frac{4}{2}}(6) + \left(\frac{1}{17}\right)(35)^{\frac{4}{2}} - \left(\frac{1}{34}\right)$$

$$Sm(\varphi, 4) = (6)\left((35)^0 + (35)^1 + (35)^2\right) + (12)\left((35)^0 + (35)^1\right)$$
$$+ \left(\frac{1}{17}\right)\left((35)^0 + (35)^1 + (35)^2\right) + \left(\frac{1}{17}\right)\left((35)^0 + (35)^1\right)$$
$$- \left(\frac{1}{34}\right)(4+1)$$

$$Sm(\varphi, n) = \left((6)\sum_{k=0}^{\left(\left(\frac{n}{2}\right)\right)}\left((35)^k\right) + (12)\sum_{k=0}^{\left(\left(\frac{(n-1)}{2}\right)\right)}\left((35)^k\right)\right) + \left(\frac{1}{17}\right)\left(\sum_{k=0}^{\left(\left(\frac{n}{2}\right)\right)}\left((35)^k\right) + \sum_{k=0}^{\left(\left(\frac{(n-1)}{2}\right)\right)}\left((35)^k\right)\right) - \left(\frac{1}{34}\right)(n+1)$$

$$= \left((6)+\left(\frac{1}{17}\right)\right)\sum_{k=0}^{\left(\left(\frac{n}{2}\right)\right)}\left((35)^k\right) + \left((12)+\left(\frac{1}{17}\right)\right)\sum_{k=0}^{\left(\left(\frac{(n-1)}{2}\right)\right)}\left((35)^k\right) - \left(\frac{1}{34}\right)(n+1)$$

$$= \left(\frac{103}{17}\right)\sum_{k=0}^{\left(\left(\frac{n}{2}\right)\right)}\left((35)^k\right) + \left(\frac{205}{17}\right)\sum_{k=0}^{\left(\left(\frac{(n-1)}{2}\right)\right)}\left((35)^k\right) - \left(\frac{1}{34}\right)(n+1)$$

$$= \left(\frac{103}{17}\right)\left(\frac{(35)^{\left(\left(\left(\frac{n}{2}\right)\right)+1\right)}-1}{34}\right) + \left(\frac{205}{17}\right)\left(\frac{(35)^{\left(\left(\left(\frac{(n-1)}{2}\right)\right)+1\right)}-1}{34}\right) - \left(\frac{1}{34}\right)(n+1)$$

6 (3a) Consider the following recurrence relation for $\varphi(n)$ and **answer the questions**:

$$\begin{cases} \varphi(0) \coloneqq c \\ \forall n \in \mathbb{N} \qquad \varphi(n+1) \coloneqq \big(\varphi(n)\big)b + d \end{cases}$$

Show calculations on pages that follow, numbered according to instructions.

Rewrite the recursion using your values of $a, b, c,$ and d.

$$\begin{cases} \varphi(0) \coloneqq (4) \\ \forall n \in \mathbb{N} \qquad \varphi(n+1) \coloneqq \big(\varphi(n)\big)(7) + (5) \end{cases}$$

Show computations on pages that follow, numbered as per instructions, for making a conjecture:

2 (i) Conjecture:

$$\forall n \in \mathbb{N} \qquad \varphi(n) = (4)(7)^n + (5)\left(\frac{7^n - 1}{6}\right)$$

4 (i) Prove your conjecture by induction:

3a.i.0

$$\begin{array}{rcl}
\varphi(n+1) & = & (\varphi(n))(7) + 5 \\
\hline
\varphi(0+1) & = & (\varphi(0))(7) + 5 \\
 & = & (4+0+0)(7) + 5 \\
\hline
\varphi(1) & = & (4)(7)+5
\end{array} \quad \langle n \leftarrow 0 \rangle$$

3a.i.1

$$\begin{array}{rcl}
\varphi(n+1) & = & (\varphi(n))(7) + 5 \\
\hline
\varphi(1+1) & = & (\varphi(1))(7) + 5 \\
 & = & ((4)(7)+5+0)(7) + 5 \\
\hline
\varphi(2) & = & (4)(7)(7)+(5)(7)+5
\end{array} \quad \langle n \leftarrow 1 \rangle$$

3a.i.2

$$\begin{array}{rcl}
\varphi(n+1) & = & (\varphi(n))(7) + 5 \\
\hline
\varphi(2+1) & = & (\varphi(2))(7) + 5 \\
 & = & ((4)(7)(7)+(5)(7)+5)(7) + 5 \\
\hline
\varphi(3) & = & (4)(7)(7)(7)+(5)(7)(7)+(5)(7)+5
\end{array} \quad \langle n \leftarrow 2 \rangle$$

3a.i.3

$$\begin{array}{rcl}
\varphi(0) & = & 4 \\
\varphi(1) & = & (4)(7)+5 \\
\varphi(2) & = & (4)(7)(7)+(5)(7)+5 \\
\varphi(3) & = & (4)(7)(7)(7)+(5)(7)(7)+(5)(7)+5 \\
\hline
\varphi(n) & = & (4)(7)^n+(5)\left(\frac{7^n-1}{6}\right)
\end{array}$$

3a.ii.0

$$\forall n \in \mathbb{N}\left(P(n) := \left(\varphi(n) = (4)(7)^n+(5)\left(\frac{7^n-1}{6}\right)\right)\right)$$

3a.ii.1

Bc

$$\begin{array}{rcl}
\text{LS} & & \text{RS} \\
= \varphi(0) & & = (4)(7)^0+(5)\left(\frac{7^0-1}{6}\right) \\
= 4 & & = (4)(1)+(5)\left(\frac{1-1}{6}\right) \\
 & & = 4+(5)\left(\frac{0}{6}\right) \\
 & & = 4+(5)(0) \\
 & & = 4+0 \\
 & & = 4 \\
\hline
\multicolumn{3}{c}{\text{LS} = 4 = \text{RS}}
\end{array}$$

Bc

3a.ii.2

Rcs

$$\begin{array}{rcl}
\multicolumn{3}{c}{P(0)\ P(1) \ldots P(n)} \\
\hline
\multicolumn{3}{c}{\text{LS}} \\
= & \varphi(n+1) & \\
= & (\varphi(n))(7) + 5 & \\
= & \left((4)(7)^n+(5)\left(\frac{7^n-1}{6}\right)\right)(7) + 5 & \\
= & (4)(7)^{n+1}+(5)\left(\frac{7^{n+1}-7}{6}\right) + 5 & \\
= & (4)(7)^{n+1}+(5)\left((1)\left(\frac{7^{n+1}-7}{6}\right)+1\right) & \\
= & (4)(7)^{n+1}+(5)\left((1)\left(\frac{7^{n+1}-1}{6}\right)\right) & \\
= & (4)(7)^{n+1}+(5)\left(\frac{7^{n+1}-1}{6}\right) & \\
\hline
\multicolumn{3}{c}{P(n+1)}
\end{array}$$

Rcs

3a.ii.3

$$\frac{\forall n \in \mathbb{N}\left(P(0)\ \left(\frac{P(0)\ P(1) \ldots P(n)}{P(n+1)}\right)\right)}{\forall n \in \mathbb{N}(P(n))}$$

$$\forall n \in \mathbb{N}\left(\varphi(n) = (4)(7)^n+(5)\left(\frac{7^n-1}{6}\right)\right)$$

6 (3b) Prove, following the method shown in class, that: $(abcd)^{\frac{1}{abcd+1}}$ **is irrational**, that is: ~~*Rtl*~~ $\left((abcd)^{\frac{1}{abcd+1}} \right)$.

First rewrite the problem using your values of a, b, c and d.

My Problem:

$$\text{~~Rtl~~}\left(((2)(7)(4)(5))^{\frac{1}{(2)(7)(4)(5)+1}}\right)$$

that is: ~~*Rtl*~~ $\left((280)^{\frac{1}{(281)}}\right)$

Start proof on this side and continue on pages that follow, numbered according to instructions.

SPf

$$\neg\left(\text{~~Rtl~~}\left((280)^{\left(\frac{1}{281}\right)}\right)\right)$$

$$\mathrm{Rtl}\left((280)^{\left(\frac{1}{281}\right)}\right)$$

$$\mathrm{Rtl}\left(\sqrt[281]{280}\right)$$

$$\exists!\, p,q \in \mathbb{N}\left(\left(\gcd(p,q)=1\right) \wedge \left(\sqrt[281]{280}=\frac{p}{q}\right)\right)$$

$$\left(\gcd(p,q)=1\right) \qquad \left(\sqrt[281]{280}=\frac{p}{q}\right)$$

$$\sqrt[281]{280}=\frac{p}{q}$$
$$(\ )^{281} \qquad (\ )^{281}$$

$$\left(\sqrt[281]{280}\right)^{281}=\left(\frac{p}{q}\right)^{281}$$

$$280=\frac{p^{281}}{q^{281}}$$
$$\times q^{281} \qquad \times q^{281}$$

$$280q^{281} = p^{281}$$

$$(7)(5)(2)(2)(2)(q^{281}) = p^{281}$$

$$\text{prm}(2) \quad 2 \mid p^{281}$$

$$2 \mid p$$

$$\exists! r \in \mathbb{N}\,(p = 2r)$$

$$2q^{281} = (2r)^{281}$$

$$\begin{array}{cc} 2q^{281} & = 2^{281}r^{281} \\ \div 2 & \quad \div 2 \end{array}$$

$$q^{281} = 2^{280}r^{281}$$

$$2^{280}r^{281} = q^{281}$$

$$\text{prm}(2) \quad 2 \mid q^{281}$$

$$2 \mid q$$

$$2 \mid p \quad 2 \mid q$$

$$2 \mid (\gcd(p,q))$$

$$2 \mid (\gcd(p,q)) \qquad \gcd(p,q) = 1$$

$$2 \mid 1$$

$$\perp$$

$$\neg\left(\neg\left(\cancel{Rtt}\left((280)^{\left(\frac{1}{281}\right)}\right)\right)\right)$$

$$\cancel{Rtt}\left((280)^{\left(\frac{1}{281}\right)}\right)$$

E Pf